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# Practical 8: K-Means Clustering
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# 1. Import libraries
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

# 2. Load the dataset
data = load_iris(as_frame=True)

X = data.data

print("Dataset: Iris Dataset")
print("Number of records:", X.shape[0])
print("Number of features:", X.shape[1])

print("\nFirst five rows:")
print(X.head())

# 3. Scale the data
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# 4. Find optimal K using Elbow Method
wcss = []
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for k in range(1, 11):

    kmeans = KMeans(
        n_clusters=k,
        random_state=42,
        n_init=10
    )

    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

# 5. Plot Elbow graph
plt.figure(figsize=(8, 5))

plt.plot(range(1, 11), wcss, marker='o')

plt.xlabel("Number of Clusters (K)")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.grid(True)
plt.show()

# 6. Apply K-Means with 3 clusters
kmeans = KMeans(
    n_clusters=3,
    random_state=42,
    n_init=10
)

clusters = kmeans.fit_predict(X_scaled)
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# 7. Add cluster labels to dataset
df = X.copy()
df["Cluster"] = clusters

print("\nClustered Data:")
print(df.head(10))

# 8. Calculate Silhouette Score
score = silhouette_score(X_scaled, clusters)

print("\nSilhouette Score:", round(score, 4))

# 9. Display cluster distribution
print("\nCluster Distribution:")
print(df["Cluster"].value_counts().sort_index())

# 10. Visualize clusters
plt.figure(figsize=(8, 5))

plt.scatter(
    X_scaled[:, 0],
    X_scaled[:, 1],
    c=clusters,
    cmap="viridis",
    s=50
)

plt.xlabel("Sepal Length (Scaled)")
plt.ylabel("Sepal Width (Scaled)")
plt.title("K-Means Clustering")
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plt.colorbar(label="Cluster")
plt.grid(True)
plt.show()

# 11. Visualize cluster distribution
df["Cluster"].value_counts().sort_index().plot(
   kind="bar",
   figsize=(8, 5)
)

plt.xlabel("Cluster")
plt.ylabel("Number of Data Points")
plt.title("K-Means Cluster Distribution")
plt.grid(axis="y", linestyle="--", alpha=0.5)
plt.tight_layout()
plt.show()
```